EC3SD03: Numerical Analysis

Lecture 5
Solution of Nonlinear Equations

Open Methods

Fixed-Point Iteration

- In this method the function is rearranged so that x is on the left-hand side of the equation:

$$f(x) = 0 \quad \Rightarrow \quad g(x) = x$$

$$x_k = g(x_{k-1}) \quad x_o \text{ is given,} \quad k = 1, 2, \ldots$$

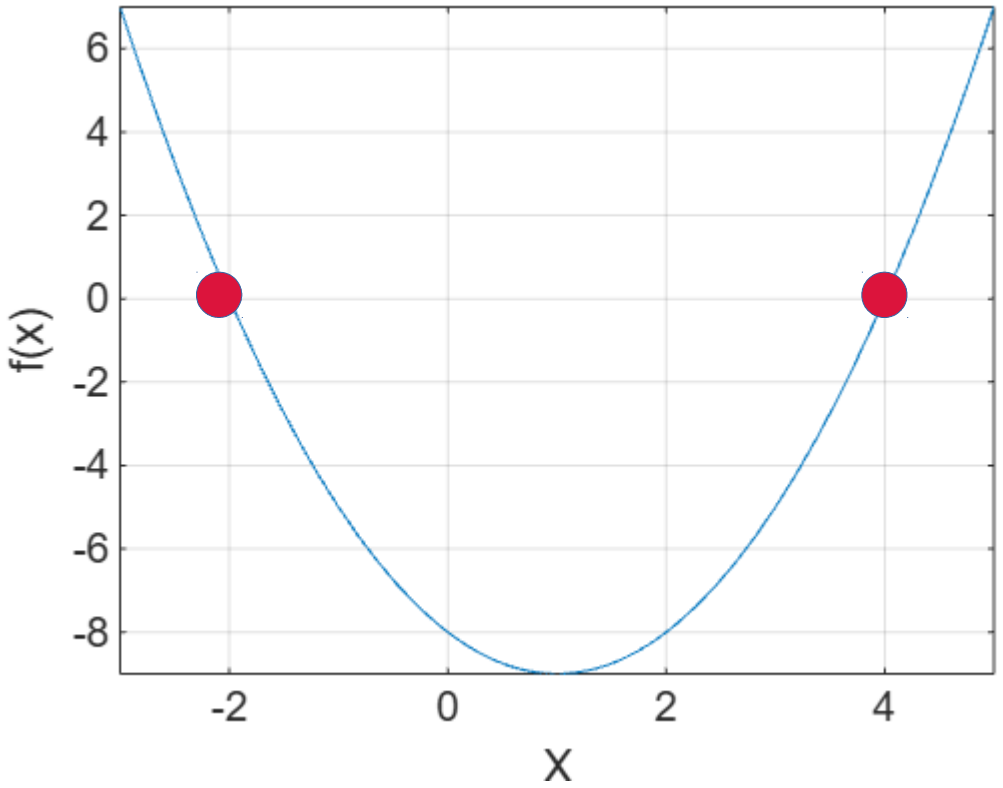
Ex: Find the roots of the equation: $f(x) = x^2 - 2x - 8 = 0$

Approximate error ε_a not greater than 10^{-4}.

Read Chapter 6 (6.1 6.2 of the textbook)

Fixed-Point Iteration: Example

$$f(x) = x^2 - 2x - 8 = 0$$



Fixed-Point Iteration

1- Rewrite the equation as : $x = g(x)$

$$x = \frac{x^2 - 8}{2}$$

or

$$x = \sqrt{2x + 8}$$

2- Select initial guess $x_0 = 2$, and calculate x_1 using the formula:

$$x_{i+1} = \frac{x_i^2 - 8}{2}$$

$$x_1 = \frac{x_0^2 - 8}{2} = \frac{4 - 8}{2} = -2$$

Fixed-Point Iteration

- Compute the approximate error:

$$\left|\epsilon_a\right| = \left|\frac{x_m^{new} - x_m^{old}}{x_m^{new}}\right| \times 100$$

$$\left|\epsilon_a\right| = \left|\frac{(-2)-(2)}{(-2)}\right| \times 100 = 200$$

- Calculate the new value of x_2

$$x_2 = \frac{x_1^2 - 8}{2} = \frac{(-2)^2 - 8}{2} = -2$$

Fixed-Point Iteration

- Next we calculate the new approximate error

$$\left|\epsilon_a\right| = \left|\frac{(-2)-(-2)}{-2}\right| \times 100 = 0$$

Since the error is 0, so the root of the equation is -2.

Fixed-Point Iteration

- However, if we choose another initial guess, for example $x_0 = 1$. The procedure is summarized in the Table

$$x_{i+1} = \frac{x_i^2 - 8}{2}$$

Iteration no.	x_i	x_{i+1}	$\lvert \in_a \rvert$
1	1	-3.5000	128.57 %
2	-3.5000	2.1250	264.71 %

Fixed-Point Iteration

It is clear that, the iterations diverges (do not converge) from the root .

Iteration no.	x_i	x_{i+1}	$\lvert \in_a \rvert$
1	1	-3.5000	128.57
2	-3.5000	2.1250	264.71
3	2.1250	-1.7422	221.97
4	-1.7422	-2.4824	29.82
5	-2.4824	-0.9189	17.016
6	-0.9189	-3.5778	74.32
7	-3.5778	2.4005	249.05
8	2.4005	-1.1189	314.55
9			

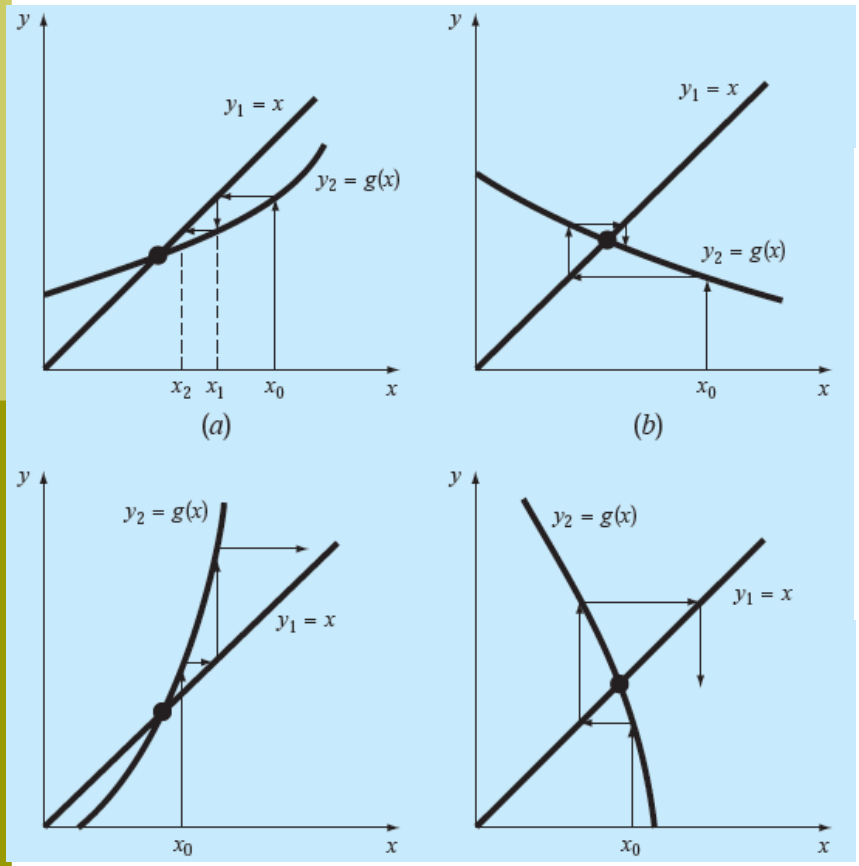
Fixed-Point Iteration

- Note that if we use the same initial guess ($x_0 = 1$) but with the formula $x = \sqrt{2x + 8}$

Iteration no.	x_i	x_{i+1}	$\lvert\epsilon_a\rvert$ %
1	1	3.1623	68.38
2	3.1623	3.7848	16.45
3	3.7848	3.9458	4.08
4	3.9458	3.9864	1.02
5	3.9864	3.9966	0.25
6	3.9966	3.9992	6.3629e-002
7	3.9992	3.9998	1.5907e-002
8	3.9998	3.9999	3.9767e-003
9	3.9999	4.0000	3.9767e-003
10	4.0000	4.0000	9.9417e-006

Fixed-Point Iteration

- Fixed-point methods may sometimes "converge", *depending on the starting point (initial guess)* and how the function behaves.



- Note that convergence occurs when $|g'(x)| < 1$.
- In other words, convergence occurs if the magnitude of the slope of $g(x)$ is less than the slope of the line $f(x) = x$.

$$g'(x) < f'(x)$$

Newton-Raphson Method

Newton-Raphson Method
(Also known as Newton's Method)

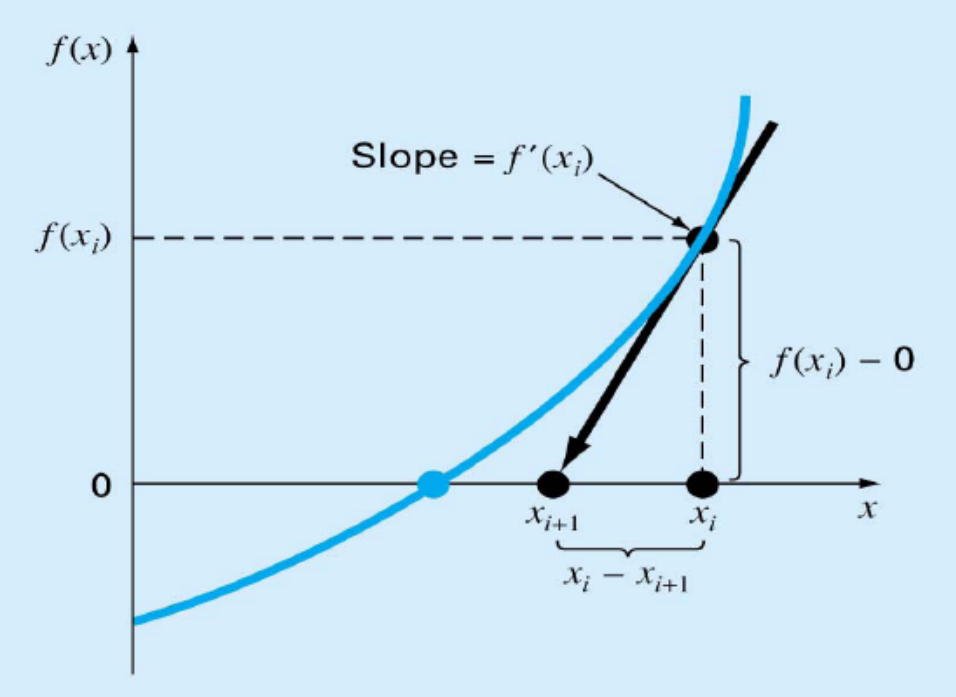
Given an initial guess of the root x_0, Newton-Raphson method uses information about the function and its derivative at that point to find a better guess of the root.

Assumptions:

- $f(x)$ is continuous and the first derivative is known
- An initial guess x_0 such that $f'(x_0) \neq 0$ is given

Newton Raphson Method
- Graphical Depiction -

- If the initial guess at the root is x_i, then a tangent to the function of x_i that is $f'(x_i)$ is extrapolated down to the *x-axis* to provide an estimate of the root at x_{i+1}.



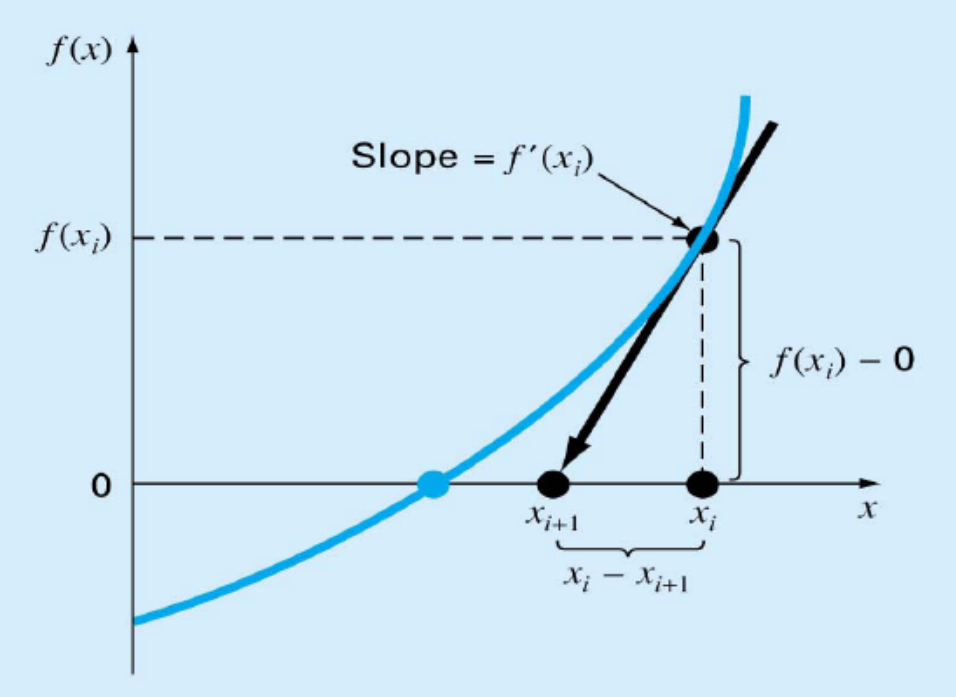
How to Calculate x_{i+1} ?

Newton Raphson Method
- Graphical Depiction -

$$f'(x_i) = \frac{f(x_i) - 0}{(x_i - x_{i+1})}$$

rearrange to obtain :

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$



Derivation of Newton's Method

Given: x_i an initial guess of the root of $f(x) = 0$

Question: How do we obtain a better estimate x_{i+1}?

Taylor Therorem: $f(x+h) \approx f(x) + f'(x)h$

Find h such that $f(x+h) = 0$.

$$\Rightarrow h \approx -\frac{f(x)}{f'(x)}$$

Newton – Raphson Formula

A new guess of the root: $x_{i+1} = x_i - \dfrac{f(x_i)}{f'(x_i)}$

Newton Raphson Method

The steps of the Newton-Raphson method to find the root of an equation $f(x) = 0$ are

1. Evaluate $f'(x)$ symbolically.
2. Use an initial guess of the root, x_i , to estimate the new value of the root, x_{i+1} , as

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

3. Find the absolute relative approximate error

$$\epsilon_a = \left| \frac{x_{i+1} - x_i}{x_{i+1}} \right| \times 100$$

4. Compare the absolute relative approximate error with the pre-specified relative error tolerance ϵ_s, . If $\epsilon_a > \epsilon_s$, then go to Step 2, else stop the algorithm. Also, check if the number of iterations has exceeded the maximum number of iterations allowed. If so, one needs to terminate the algorithm and notify the user.

Example

Find a zero of the function $f(x) = x^3 - 2x^2 + x - 3$, $x_0 = 4$

$f'(x) = 3x^2 - 4x + 1$

Iteration 1: $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 4 - \frac{33}{33} = 3$

Iteration 2: $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 3 - \frac{9}{16} = 2.4375$

Iteration 3: $x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 2.4375 - \frac{2.0369}{9.0742} = 2.2130$

Example

#	x_k	$f(x_k)$	$f'(x_k)$	x_{k+1}	$\lvert x_{k+1} - x_k \rvert$	$\lvert \epsilon_a \rvert$
0	**4**	**33**	**33**	**3**	1	33.3%
1	**3**	**9**	**16**	**2.4375**	0.5625	23 %
2	**2.4375**	**2.0369**	**9.0742**	**2.2130**	0.2245	9.2%
3	**2.2130**	**0.2564**	**6.8404**	**2.1756**	0.0384	1.7%
4	**2.1756**	**0.0065**	**6.4969**	**2.1746**	0.0010	0.046

Convergence Analysis
Remarks

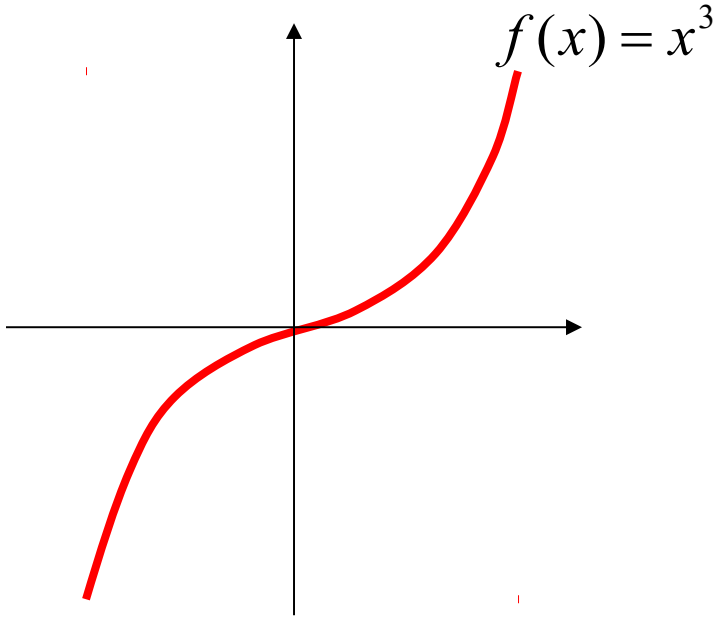
When the guess is close enough to a simple root of the function then Newton's method is guaranteed to converge quadratically.

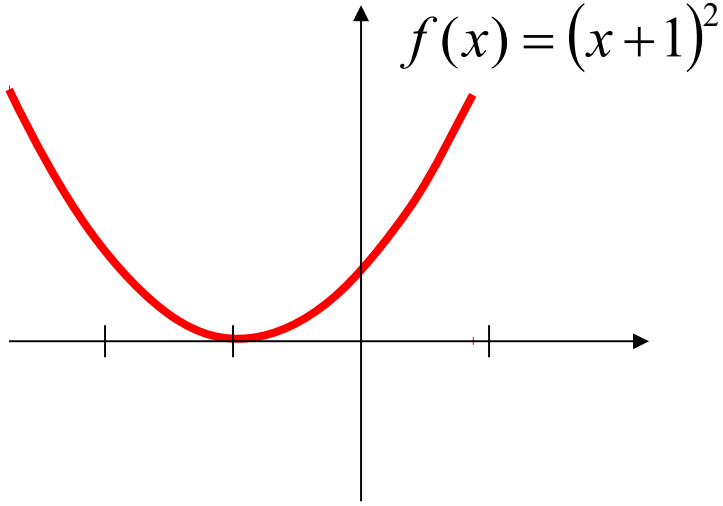
Quadratic convergence means that the number of correct digits is nearly doubled at each iteration.

Problems with Newton's Method

- If the initial guess of the root is far from the root the method may not converge.
- Newton's method converges linearly near multiple zeros { ***f(r) = f'(r) =0*** }. In such a case, modified algorithms can be used to regain the quadratic convergence.

Multiple Roots





f(x) has three zeros at x = 0

f(x) has two zeros at $x = -1$

Example 2

Use Newton-Raphson method to find the root of the function

$$x^3 - 0.165x^2 + 3.993\times10^{-4} = 0$$

$$f(x) = x^3 - 0.165x^2 + 3.993\times10^{-4}$$

$$f'(x) = 3x^2 - 0.33x$$

Let us assume the initial guess of the root of $f(x) = 0$ is $x_0 = 0.05\,\text{m}$.

- **Iteration 1** : The estimate of the root is

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= 0.05 - \frac{(0.05)^3 - 0.165(0.05)^2 + 3.993\times10^{-4}}{3(0.05)^2 - 0.33(0.05)}$$

$$= 0.05 - \frac{1.118\times10^{-4}}{-9\times10^{-3}} = 0.05 - (-0.01242) = 0.06242$$

Example 2:

The absolute relative approximate error $|\epsilon_a|$ at the end of Iteration 1 is

$$|\epsilon_a| = \left|\frac{x_1 - x_0}{x_1}\right| \times 100$$

$$= \left|\frac{0.06242 - 0.05}{0.06242}\right| \times 100$$

$$= 19.90\%$$

The number of significant digits at least correct is 0, as you need an absolute relative approximate error of 5% or less for at least one significant digit to be correct in your result.

- **Iteration 2:** The estimate of the root is

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 0.06242 - \frac{(0.06242)^3 - 0.165(0.06242)^2 + 3.993\times10^{-4}}{3(0.06242)^2 - 0.33(0.06242)} = 0.06238$$

Example 2:

The absolute relative approximate error $|\epsilon_a|$ at the end of Iteration 2 is

$$|\epsilon_a| = \left|\frac{x_2 - x_1}{x_2}\right| \times 100 = \left|\frac{0.06238 - 0.06242}{0.06238}\right| \times 100 = 0.0716\%$$

The maximum value of m for which $|\epsilon_a| \le 0.5 \times 10^{2-m}$ is 2.84. Hence, the number of significant digits at least correct in the answer is 2.

Iteration 3: The estimate of the root is

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$= 0.06238 - \frac{(0.06238)^3 - 0.165(0.06238)^2 + 3.993 \times 10^{-4}}{3(0.06238)^2 - 0.33(0.06238)} = 0.06238$$

The absolute relative approximate error $|\epsilon_a|$ at the end of Iteration 3 is

$$|\epsilon_a| = \left|\frac{0.06238 - 0.06238}{0.06238}\right| \times 100 = 0$$

The number of significant digits at least correct is 4, as only 4 significant digits are carried through in all the calculations .

Drawbacks of the Newton-Raphson Method

1. **Divergence at inflection points:** If the selection of a guess or an iterated value turns out to be close to the inflection point of f(x), that is, near where f''(x)=0, the roots start to diverge away from the true root.

2. **Division by zero or near zero:** The formula of Newton-Raphson method is

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Consequently if an iteration value, is such that $f'(x_i) \cong 0$, then one can face division by zero or a near-zero number. This will give a large magnitude for the next value, x_i+1. An example is finding the root of the equation

$$f(x) = x^3 - 0.03x^2 + 2.4 \times 10^{-6}$$

in which case

$$f'(x) = 3x^2 - 0.06x$$

For $x_0 = 0$ or $x_0 = 0.02$, division by zero occurs. For an initial guess close to 0.02, of $x_0 = 0.01999$, even after 9 iterations (next Table), the Newton-Raphson method is not converging.

Table 5

Iteration Number	x_i	$\lvert\varepsilon_a\rvert$	$f(x_i)$
0	0.019990		-1.6000×10^{-6}
1	-2.6480	100.75	-18.778
2	-1.7620	50.282	-5.5638
3	-1.1714	50.422	-1.6485
4	-0.77765	50.632	-0.48842
5	-0.51518	50.946	-0.14470
6	-0.34025	51.413	-0.042862
7	-0.22369	52.107	-0.012692
8	-0.14608	53.127	-0.0037553
9	-0.094490	54.602	-0.0011091

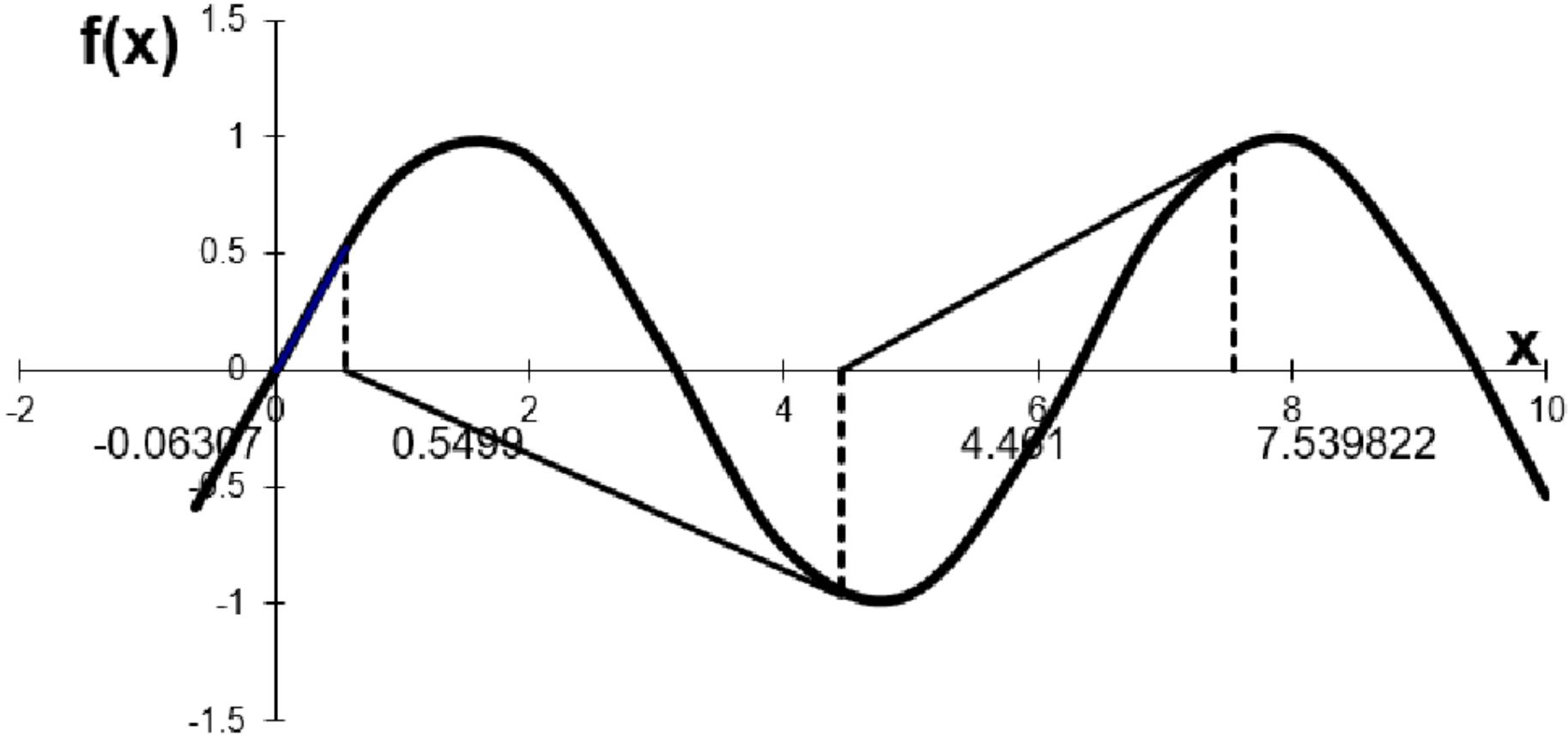
3. Root jumping: In some case where the function $f(x)$ is oscillating and has a number of roots, one may choose an initial guess close to a root. However, the guesses may jump and converge to some other root. For example for solving the equation

$$\sin x = 0$$

if you choose $x_0 = 2.4\pi = (7.539822)$ as an initial guess, it converges to the root of x=0 as shown in next Table. However, one may have chosen this an initial guess to converge to $x = 2\pi$

Table 6

Iteration Number	x_i	$f(x_i)$	$\lvert\epsilon_a\rvert$
0	7.539822	0.951	
1	4.461	-0.969	68.973
2	0.5499	0.5226	711.44
3	-0.06303	-0.06303	971.91
4	6.376×10^{-4}	8.375×10^{-4}	7.54×10^{4}
5	-1.95861×10^{-13}	-1.95861×10^{-13}	4.27×10^{10}



Root jumping from intended location of root for $f(x) = \sin x = 0$

Newton's Method(Summary)

Assumptions: $f(x),\ f'(x),\ x_0$ *are available,*

$$f'(x_0) \neq 0$$

Newton's Method new estimate:

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Problem:

$f'(x_i)$ is not available,

or difficult to obtain analytically.

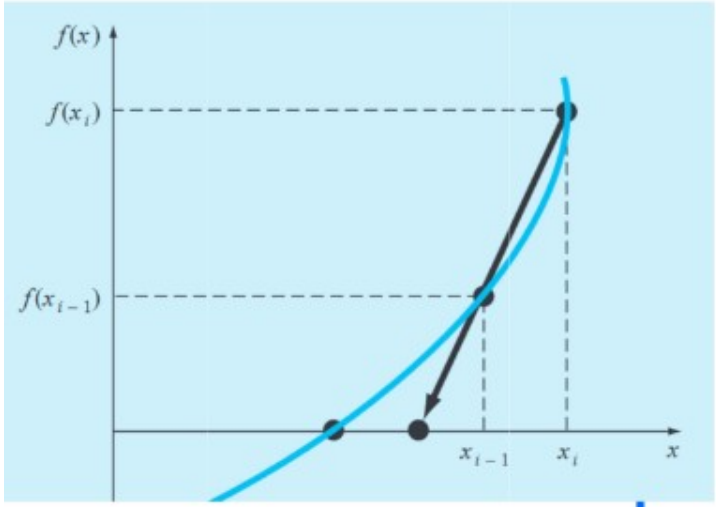
Secant Method

Secant Method

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

if x_i *and* x_{i-1} are two initial points :

$$f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{(x_i - x_{i-1})}$$

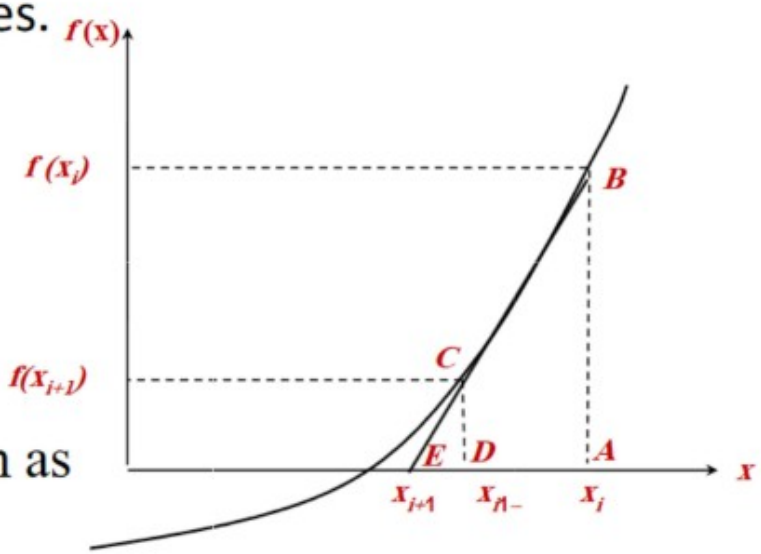


$$x_{i+1} = x_i - \frac{f(x_i)}{\dfrac{f(x_i) - f(x_{i-1})}{(x_i - x_{i-1})}} = \quad x_i - f(x_i)\frac{(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$$

From Fig. *ABE* and *DCE* are similar triangles.

$$\frac{AB}{AE} = \frac{DC}{DE}$$

$$\frac{f(x_i)}{x_i - x_{i+1}} = \frac{f(x_{i-1})}{x_{i-1} - x_{i+1}}$$



On rearranging, the secant method is given as

$$x_{i+1} = x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$$

Secant Method

Assumptions :

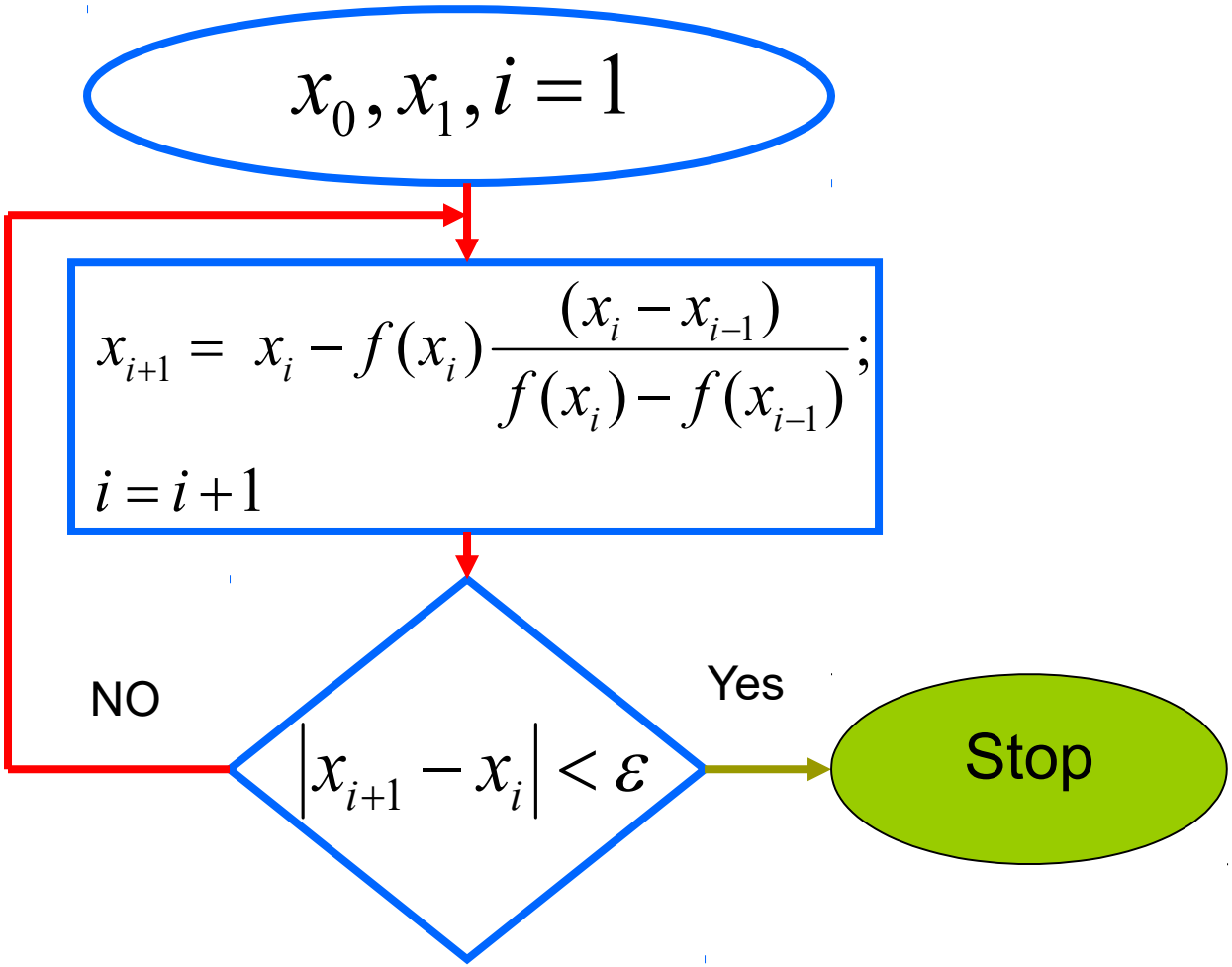
Two initial points x_i *and* x_{i-1}

such that $f(x_i) \neq f(x_{i-1})$

New estimate (Secant Method) :

$$x_{i+1} = x_i - f(x_i)\frac{(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$$

Secant Method - Flowchart



Modified Secant Method

In this modified Secant method, only one initial guess is needed :

$$f'(x_i) \approx \frac{f(x_i + \delta\, x_i) - f(x_i)}{\delta\, x_i}$$

$$x_{i+1} = x_i - \frac{f(x_i)}{\dfrac{f(x_i + \delta\, x_i) - f(x_i)}{\delta\, x_i}} = x_i - \frac{\delta\, x_i f(x_i)}{f(x_i + \delta\, x_i) - f(x_i)}$$

Problem : How to select δ ?

If not selected properly, the method may diverge.

Modified Secant Method

- The choice of a proper value for δ is not automatic.
- If δ is **too small**, the method can be swamped by **round-off error** caused by subtractive cancellation in the denominator of the root estimate equation
- If it is **too big**, the technique can become **inefficient and even divergent.**
- However, if chosen correctly, it provides a nice alternative for cases where evaluating the root of a function

Example

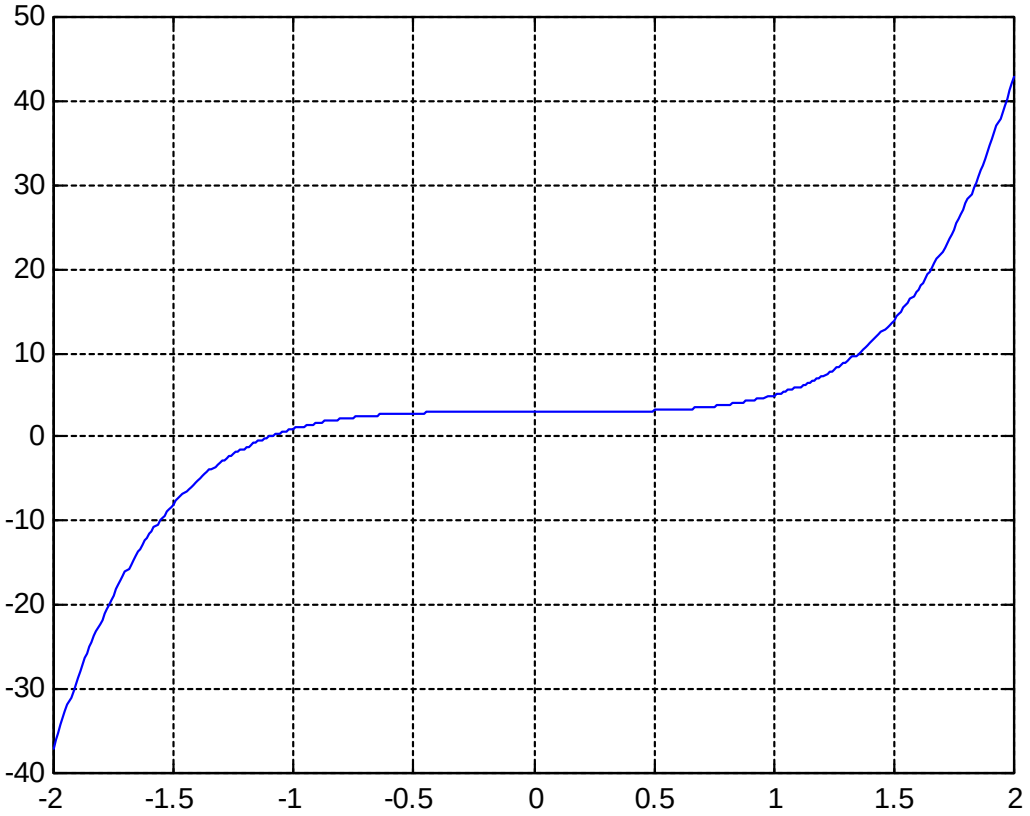
Find the roots of :

$f(x) = x^5 + x^3 + 3$

Initial points

$x_0 = -1$ *and* $x_1 = -1.1$

with error < 0.001



Example

x(i)	f(x(i))	x(i+1)	|x(i+1)-x(i)|
-1.0000	1.0000	-1.1000	0.1000
-1.1000	0.0585	-1.1062	0. 0062
-1.1062	0.0102	-1.1052	0.0009
-1.1052	0.0001	-1.1052	0.0000

Example

Find the root of the function

$$f(x) = 5 - 2e^{-x} = 0$$

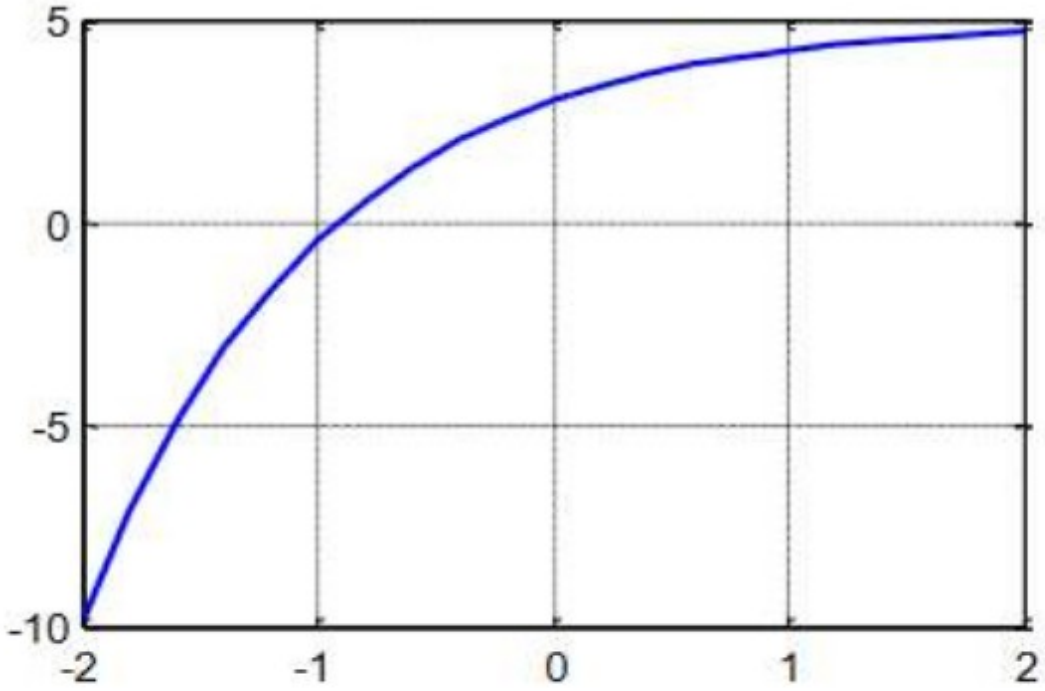
use Secant method. Plot the function to determine the initial guess.

Find the root of the function

$$f(x) = 5 - 2e^{-x} = 0$$

use Secant method. Plot the function to determine the initial guess.

- First we plot the function. From Fig. we see that the function has a root between [-2, 0]. Let us choose x0 = -2, x1 = 0.



Secant Method:

$$x_{i+1} = x_i - f(x_i)\frac{(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$$

Determine $f(x_0)$ and $f(x_1)$: $f(x_0) = 5 - 2e^{-x_0} = -9.7781$ $f(x_1) = 5 - 2e^{-x1} = 3$

Calculate x_2 using the formula:

$$x_2 = x_1 - \frac{(x_0 - x_1)f(x_1)}{f(x_0) - f(x_1)}.$$

$$x_2 = 0 - \frac{(-2 - 0) \times (3)}{(-9.7781 + 3)} = -0.4696$$

- Determine $f(x_0)$ and $f(x_1)$: $f(x_0) = 5 - 2e^{-x_0} = -9.7781$ $f(x_1) = 5 - 2e^{-x1} = 3$
- Calculate x_2 using the formula:

$$x_2 = x_1 - \frac{(x_0 - x_1) f(x_1)}{f(x_0) - f(x_1)}.$$

$$x_2 = 0 - \frac{(-2 - 0) \times (3)}{(-9.7781 + 3)} = -0.4696$$

- Determine $f(x_0)$ and $f(x_1)$: $f(x_0) = 5 - 2e^{-x_0} = -9.7781$ $f(x_1) = 5 - 2e^{-x1} = 3$
- Calculate x_2 using the formula:

$$x_2 = x_1 - \frac{(x_0 - x_1) f(x_1)}{f(x_0) - f(x_1)}.$$

$$x_2 = 0 - \frac{(-2 - 0) \times (3)}{(-9.7781 + 3)} = -0.4696$$

Now the approximate error equals to | x2 – x1| = 0.4696.

• Now **x_0 = 0 and x_1 = - 0.4696**. **Repeat steps 2 to 4** until the approximate error is **close to zero**.

● This occurs after 7 iterations as shown in next Table.

The solution is x_2 = -0.9163.

Secant method Example

Iteration No.	x_0	x_1	$f(x_0)$	$f(x_1)$	x_2	$\lvert\varepsilon_a\rvert = \lvert x_2 - x_1 \rvert$
1	-2	0	-9.7781	3	-0.4696	0.4696
2	0	-0.4696	3	1.8014	-1.1753	0.7057
3	-0.4696	-1.1753	1.8014	-1.4782	-0.8572	0.3181
4	-1.1753	-0.8572	-1.4782	0.2869	-0.9089	0.0517
5	-0.8572	-0.9089	0.2869	0.0368	-0.9165	0.0076
6	-0.9089	-0.9165	0.0368	-0.0011	-0.9163	2.2151e-004
7	-0.9165	-0.9163	-0.0011	4.0809e-006	-0.9163	8.1609e-007

Example 2:

- Use the secant method to estimate the root of
- $f(x) = e^{-x} - x$.
- Start with initial estimates of $x_{-1} = 0$ and $x_0 = 1.0$.

the true root is 0.56714329

Example 2: $f(x) = e^{-x} - x$

First iteration:

the true root is 0.56714329

$$x_{-1} = 0 \qquad f(x_{-1}) = 1.00000$$

$$x_0 = 1 \qquad f(x_0) = -0.63212$$

$$x_1 = 1 - \frac{-0.63212(0-1)}{1-(-0.63212)} = 0.61270 \qquad \varepsilon_t = 8.0\%$$

$$\text{True percent relative error: } \varepsilon_t = \left|\frac{\text{True value} - \text{Approximation}}{\text{True value}}\right| \times 100\%$$

$$f(x) = e^{-x} - x$$

Second iteration:

$x_0 = 1$ $\quad f(x_0) = -0.63212$

$x_1 = 0.61270$ $\quad f(x_1) = -0.07081$

(Note that both estimates are now on the same side of the root.)

$$x_2 = 0.61270 - \frac{-0.07081(1 - 0.61270)}{-0.63212 - (-0.07081)} = 0.56384 \qquad \varepsilon_t = 0.58\%$$

Third iteration:

$x_1 = 0.61270$ $\quad f(x_1) = -0.07081$

$x_2 = 0.56384$ $\quad f(x_2) = 0.00518$

$$x_3 = 0.56384 - \frac{0.00518(0.61270 - 0.56384)}{-0.07081 - (-0.00518)} = 0.56717 \qquad \varepsilon_t = 0.0048\%$$

Use the modified secant method to estimate the root of $f(x) = e^{-x} - x$. Use a value of 0.01 for δ and start with x0 = 1.0. Recall that the true root is 0.56714329

First iteration:

$$x_0 = 1 \qquad f(x_0) = -0.63212$$

$$x_0 + \delta x_0 = 1.01 \qquad f(x_0 + \delta x_0) = -0.64578$$

$$x_1 = 1 - \frac{0.01(-0.63212)}{-0.64578 - (-0.63212)} = 0.537263 \qquad |\varepsilon_t| = 5.3\%$$

Second iteration:

$$x_0 = 0.537263 \qquad f(x_0) = 0.047083$$

$$x_0 + \delta x_0 = 0.542635 \qquad f(x_0 + \delta x_0) = 0.038579$$

$$x_1 = 0.537263 - \frac{0.005373(0.047083)}{0.038579 - 0.047083} = 0.56701 \qquad |\varepsilon_t| = 0.0236\%$$

Third iteration:

$$x_0 = 0.56701 \qquad f(x_0) = 0.000209$$

$$x_0 + \delta x_0 = 0.572680 \qquad f(x_0 + \delta x_0) = -0.00867$$

$$x_1 = 0.56701 - \frac{0.00567(0.000209)}{-0.00867 - 0.000209} = 0.567143 \qquad |\varepsilon_t| = 2.365 \times 10^{-5}\%$$

Finding the Root: Comparing Numerical Methods

Bracketing Methods (The Reliable Choice)

Always find a root if it is bracketed.

Bisection Method

Stability

Reduces the interval by exactly 50% per iteration without needing derivative knowledge.

False Position

Graphical Insight

Uses a straight line between points; usually faster than Bisection but can stall.

Guaranteed Convergence

Open Methods (The Efficient Choice)

Start with initial guesses, may jump away from roots.

Quadratic

Convergence Speed

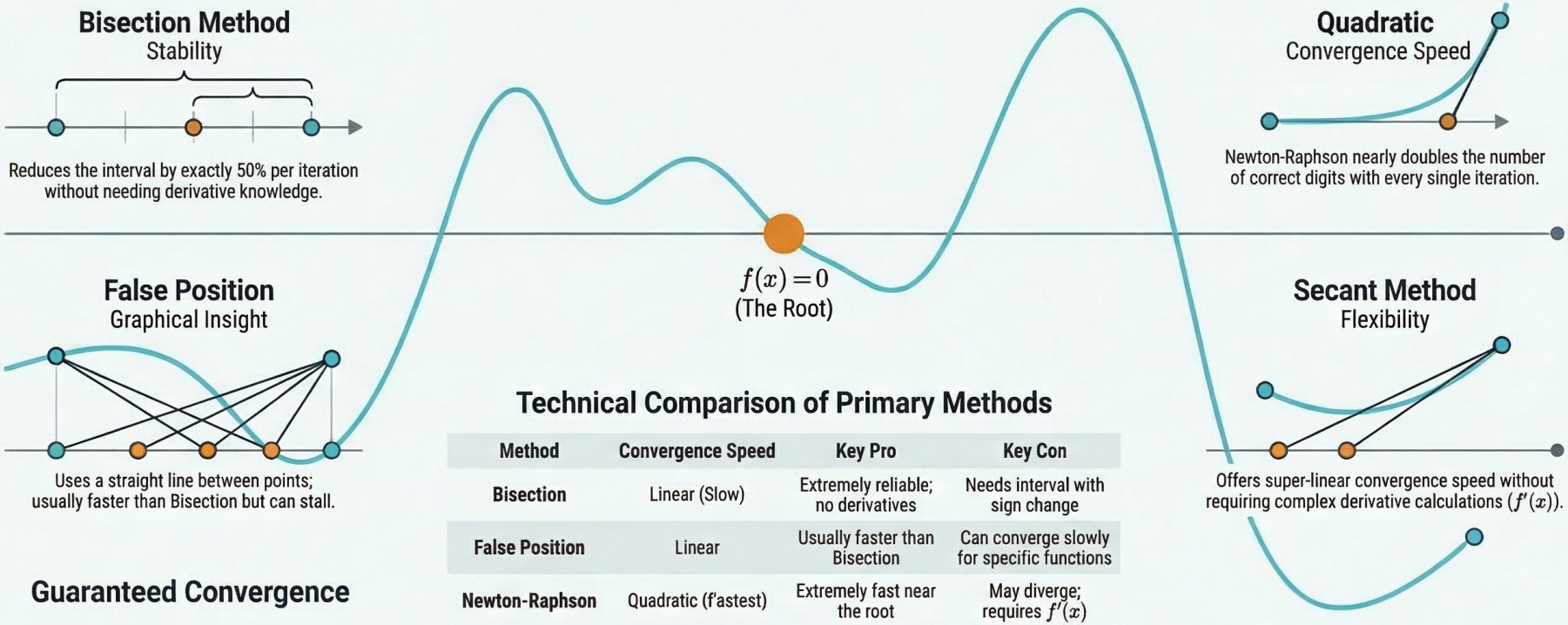
Newton-Raphson nearly doubles the number of correct digits with every single iteration.

Secant Method

Flexibility

Offers super-linear convergence speed without requiring complex derivative calculations ($f'(x)$).

The Risk of Divergence



Technical Comparison of Primary Methods

Method	Convergence Speed	Key Pro	Key Con
Bisection	Linear (Slow)	Extremely reliable; no derivatives	Needs interval with sign change
False Position	Linear	Usually faster than Bisection	Can converge slowly for specific functions
Newton-Raphson	Quadratic (f'astest)	Extremely fast near the root	May diverge; requires $f'(x)$
Secant	Super-linear	Fast; no derivatives needed	May diverge; needs two initial guesses

NotebookL